



Dept of Physics, IIT Delhi

2501-PYL1003 (Mechanics & STR)

Course Instructor: Pradipta Ghosh

Full Marks: 30, Set-A

Mid-sem exam, Date: 12-09-2025

Name:

Entry Number:

Please read the following instructions carefully

- Please write your name and entry numbers on the question papers at the given places.
- An MCQ can also be an MSQ. Explain your answer for the MCQ/MSQ.
- Please show the intermediate steps and reasoning for your answers.
- Rough work should be shown in the question paper itself.
- Approximation, truncation/round-off should be done justifiably.

1. In terms of the fundamental units (M, L, T, etc.), 'torque' has the same dimension as **2 + 1 = 3**

- ☐ a Jerk
- ☐ b Kinetic Energy
- ☐ c Angular Momentum
- ☐ d Work

2 (i). The dimension of the Gravitational constant G, in terms of the fundamental units (M, L, T, etc.) is **1 + 1 + 1 = 3**

- ☐ a $M^1 L^3 T^{-2}$
- ☐ b $M^{-2} L^3 T^{-1}$
- ☐ c $M^{-1} L^2 T^{-3}$
- ☐ d $M^{-1} L^3 T^{-2}$

2 (ii). The magnitude of the gravitational force between a massless particle and a solid metal ball of mass 1 kg, separated by a distance of 10^{-6} m, is _____ (write in terms of G).

3 (i). A toy car of mass 0.1 kg is moving in a straight line with a constant speed of 10 m/sec. Suddenly, it starts to decelerate at a rate of $0.8t$ m/secⁿ. [Neglect friction. Treat the instant of deceleration as a reference starting point]. The value of n is **1 + 1 + 1 = 3**

- ☐ a 1
- ☐ b 2
- ☐ c -1
- ☐ d 3

3 (ii). The speed goes to zero at a time $t = m$ sec. The value of m is

- ☐ a 8
- ☐ b 5
- ☐ c 4
- ☐ d 7

4 (i). A man, standing on the floor, pushes a basketball of mass 0.5 kg in a direction parallel to the floor with a force of magnitude 10 N. Later, he repeats the same action, but this time for a cart of mass 50 kg. The ratio of resultant acceleration of the ball to the same of the cart is (neglecting floor friction, air friction, and the effect of gravity)

$$1 + \frac{1}{2} + 1 + \frac{1}{2} = 3$$

- ☐ a 100
- ☐ b 25
- ☐ c 0.01
- ☐ d 0.04

4 (ii). The magnitude of the acceleration of the cart, in m/sec², will be

- ☐ a 0.20
- ☐ b 0.25
- ☐ c 0.15
- ☐ d 5.0

5. The coordinates of a point P, in the Cartesian coordinates, are given as $(x = 3, y = 4, z = -12)$. The coordinates of P, with respect to cylindrical polar and spherical polar coordinate systems, are written as (ρ, ϕ, z) and (r, θ, ϕ) , respectively. Which of the following statement(s) is(are) correct? [Hint: for spherical polar, $z = r \cos \theta$. Rounded-off to two decimal places.]

$$1 + 1 + 1 = 3$$

- ☐ a $\rho = 5, r = 13, \phi = 53.13^\circ, z = -12$
- ☐ b $\rho = 5, r = 13, \phi = 53.13^\circ, \theta = 157.38^\circ$
- ☐ c $\rho = 5, r = 13, \phi = 53.13^\circ, \theta = -22.62^\circ$
- ☐ d $\rho = 5, r = 13, \phi = 36.87^\circ, \theta = 157.38^\circ$

6. A vector $\vec{A}$, in spherical polar coordinates, is written as $\vec{A} = 3\hat{r} + 2\hat{\theta} - 25\hat{\phi}$. Here, $\hat{r}, \hat{\theta}, \hat{\phi}$ are unit vectors along r, θ , and ϕ directions, respectively. If $\hat{i}, \hat{j}, \hat{k}$ represent unit vectors along x, y, z directions in the Cartesian coordinate system, [Hint: $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, $z = r \cos \theta$, and $\hat{u} = \frac{\partial \vec{r}}{\partial u} / |\frac{\partial \vec{r}}{\partial u}|$, where $u = r, \theta, \phi$. Rounded-off to two decimal places.]

$$2 + 1\frac{1}{2} \times 2 = 5$$

- (i) calculate $\hat{k}$ in terms of $\hat{r}, \hat{\theta}, \hat{\phi}$ at a point $r = 1, \theta = 45^\circ, \phi = 90^\circ$,
- (ii) calculate $(\vec{A} \cdot \hat{k})\hat{k}$ in terms of $\hat{r}, \hat{\theta}, \hat{\phi}$ at a point $r = 1, \theta = 90^\circ, \phi = 135^\circ$,
- (iii) calculate $(\vec{A} \cdot \hat{k})\hat{k}$ in terms of $\hat{r}, \hat{\theta}, \hat{\phi}$ at a point $r = 2, \theta = 0^\circ, \phi = 90^\circ$.

7. For a planet in the solar system, the ratio of speed at the perihelion is three times the speed at the aphelion position. Calculate the eccentricity (ϵ) of the concerned orbit. [Orbit trajectory is given as $r = p/(1 + \epsilon \cos \theta)$, with p being the semi-latus rectum.]

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8. In the given system of pulleys, as shown below, a force of magnitude 10 N is needed to maintain the system at equilibrium, i.e., no net acceleration of the block B. Calculate (i) the mass (M) of the block B, and (ii) the tension (T) in the topmost cable that is connecting pulley number 1 to the ceiling. (iii) Draw the properly scaled free-body diagram. [Ignore masses of the pulleys and the connecting cables. Also, ignore air resistance. Take acceleration due to gravity as 10 m/sec² acting downward.]

$$1\frac{1}{2} \times 2 + 2 = 5$$

